

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

TABLEAU DASHBOARD PROJECT

Hotel Booking Revenue & Guest Segmentation Dashboard

Student Name	Ishita Patel
Roll No.	31
Class / Batch	TYIT /02
Subject	Data Visualization with Power BI and Tableau
Academic Year	2026-2027

Submitted to: Prof. Sumit Tripathi

1. Project Title

Hotel Booking Revenue & Guest Segmentation Dashboard Using Tableau

2. Objective

The main objective of this project is to create an interactive Hotel Booking Revenue & Guest Segmentation Dashboard using Tableau. The dashboard analyzes hotel booking data to understand revenue, profit, guest trends, room-type performance, regional performance, and customer segments through KPIs, bar charts, line charts, maps, scatter plots, and interactive filters.

3. Project Description

This project uses Tableau to create an interactive Hotel Booking Revenue & Guest Segmentation Dashboard. The dashboard analyzes hotel booking data to provide meaningful insights into revenue, profit, room types, regions, booking trends, and customer segments. It includes KPI cards, bar charts, a line chart, a city map, a scatter plot, and customer segmentation. Interactive filters allow users to explore the data based on different categories and time periods, making it easier to understand hotel performance and customer behavior.

4. Dataset Details

Parameter	Details
Dataset Name	Hotel Booking Revenue & Guest Segmentation Dataset
Source	CSV Dataset
Number of Records	800
Important Fields	Booking ID, Customer ID, Booking Date, Check-in Date, Hotel, City, Region, Room Type, Booking Category, Booking Channel, Guests, Nights, Room Rate, Discount (%), Revenue, Cost, Profit, Customer Age, Customer Gender, Payment Method, Customer Rating, Booking Status

5. Tools & Technologies Used

- Tableau Public
- CSV Dataset
- Tableau Calculated Fields
- Interactive Charts and Filters
- Data Visualization and Dashboarding

6. Tableau Worksheets / Charts

Visualization	Purpose
KPI Cards	Display Total Revenue, Total Profit and Total Guests.
Line Chart	Shows revenue trend over time.
Bar Chart – Revenue by Room Type	Compares revenue across different room types.

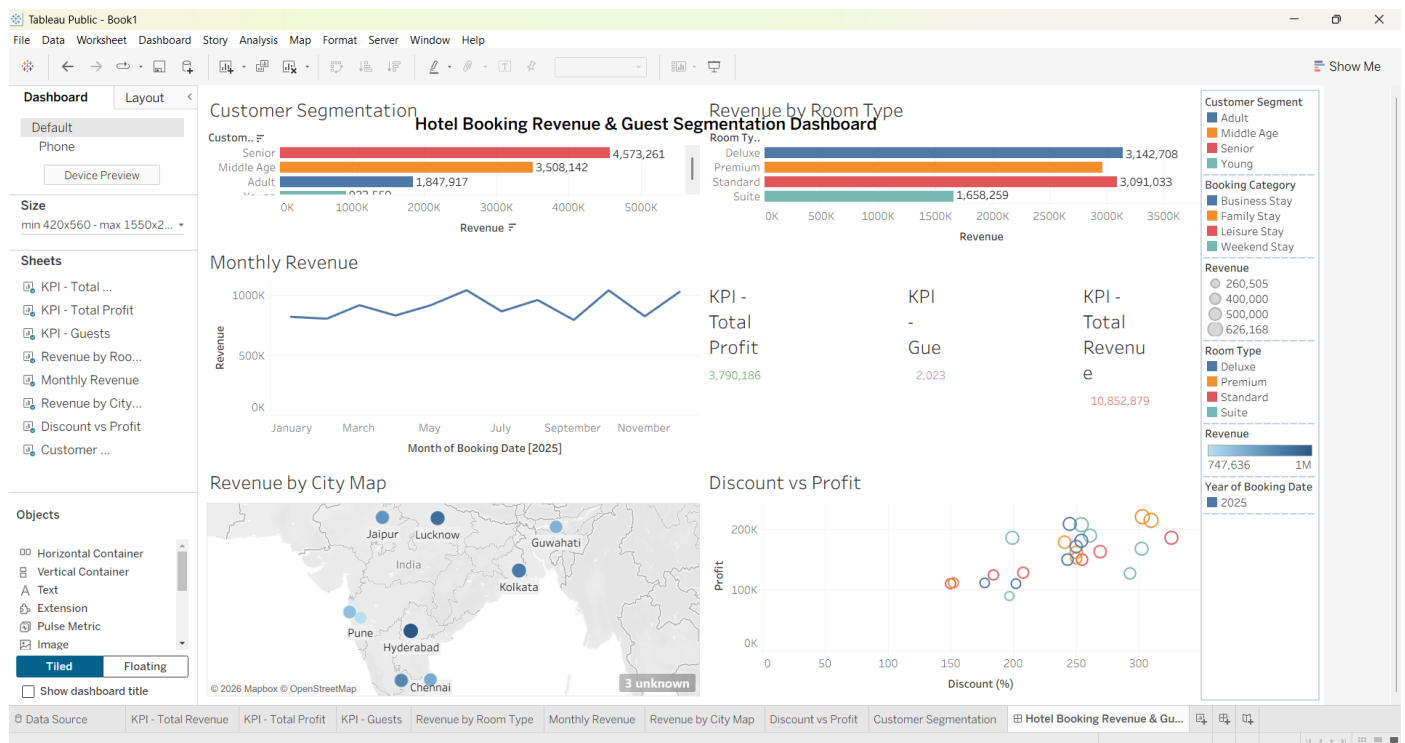
Map Visualization – Revenue by City	Shows revenue distribution by city/location.
Scatter Plot – Discount vs Profit	Shows the relationship between discount percentage and profit.
Customer Segmentation Chart	Shows revenue performance across different customer age groups.
Filter Chart	Allows interactive analysis by Year, Room Type, Region and Hotel.

7. Calculated Fields

The following calculated fields can be used in the Tableau dashboard:

1. **Total Revenue** : $\text{SUM}([\text{Revenue}])$
2. **Total Profit** : $\text{SUM}([\text{Profit}])$
3. **Profit Ratio** : $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Revenue}])$
4. **Average Revenue** : $\text{AVG}([\text{Revenue}])$
5. **Customer Segment** : IF [Customer Age] <= 25 THEN "Young" ELSEIF [Customer Age] <= 35 THEN "Adult" ELSEIF [Customer Age] <= 50 THEN "Middle Age" ELSE "Senior" END

8. Dashboard Screenshot



9. Key Insights

1. The dataset contains 800 hotel booking records, generating total revenue of approximately ₹10.85 million and total profit of approximately ₹3.79 million.
2. South Region generated the highest revenue of approximately ₹2.93 million.
3. BlueHarbor was the highest-performing hotel, generating approximately ₹2.04 million in revenue.
4. Deluxe Rooms generated the highest profit of approximately ₹1.10 million.
5. The Senior customer segment generated the highest revenue of approximately ₹4.57 million.

10. Video Demonstration

Video Link: <https://youtu.be/i97XqoyvkbM>

11. GitHub Repository

GitHub Link: <https://github.com/ishipatelmvlu-sys/Tableau>

Uploaded Files:

- **Tableau workbook (.twb / .twbx):** <https://github.com/ishipatelmvlu-sys/Tableau/blob/main/T031.twbx>
- **Dataset (.csv / .xlsx):** [https://github.com/ishipatelmvlu-sys/Tableau/blob/main/Hotel_Booking_Revenue_Guest_Segmentation_800_Rows%20\(1\).csv](https://github.com/ishipatelmvlu-sys/Tableau/blob/main/Hotel_Booking_Revenue_Guest_Segmentation_800_Rows%20(1).csv)
- **Dashboard screenshot:** <https://github.com/ishipatelmvlu-sys/Tableau/blob/main/Screenshot%202026-10-05%2011419.png>
- **README file:** <https://github.com/ishipatelmvlu-sys/Tableau/blob/main/README.md>